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Unlocking Ocean Heat: How Warm Seas Fuel Extreme Floods

Abstract

In early 2026, heavy rain again soaked Northland, Auckland, and the Coromandel in New Zealand, triggering floods and road closures. Climate scientist Kevin Trenberth explained how record ocean heat, which in 2025 rose by an amount equal to more than 200 times the world's annual electricity use, loads the atmosphere with moisture and energy, fueling atmospheric rivers, storms, and marine heat waves.

Introduction

For many people in New Zealand's upper North Island, showery days and bursts of heavy rain became all too familiar in the summer of 2026. Fresh downpours on already saturated ground triggered flood warnings and road closures. These were individual weather events, but they unfolded against unusually warm seas. Scientists have long known that sea surface temperatures influence weather: warmer water means more evaporation, altered winds, and shifting storm tracks. But the surface is only the skin of a deeper system. To understand why warm conditions persist, and why they keep feeding storms, scientists look at the heat stored throughout the upper ocean.

How Do Scientists Study This?

Since the early 2000s, profiling floats have measured temperature and salinity down to 2,000 meters in oceans worldwide. These observations allowed scientists to extend ocean heat analyses back to 1958; before then, measurements were too sparse for a global picture. An international team, including Trenberth, calculated ocean heat content, the total heat stored through the ocean's upper layers, for each year and compared 2025 with 2024. They tracked when warming reached different depths, and checked their results against physical constraints of mass, energy, water, and sea level. They also used satellite altimetry to monitor sea level, since heat and water must be tracked together.

From ocean heat to flooding

1 Record ocean heat

+23 zettajoules in 2025

2 More evaporation

Warm seas load the air

3 Atmospheric rivers

Moisture flows to land

4 Floods

Heavy rain on saturated ground

Profile the ocean

Floats measure temperature and salinity from the surface to 2,000 meters.

Add up the heat

Scientists calculate total ocean heat content and compare it year by year.

Cross-check

Results are tested against energy, water, and satellite sea level data.

What Did They Discover?

Ocean heat content reached record levels in 2025, rising about 23 zettajoules above 2024, equivalent to more than 200 times the world's annual electricity use, or enough to heat 28 billion Olympic pools from 20°C to 100°C. More than 90% of Earth's energy imbalance ends up in the ocean, making ocean heat the single best indicator of global warming. It takes about 25 years for surface heat to reach 2,000 meters. Warm seas load the air with moisture, strengthening tropical cyclones and atmospheric rivers. In 2025, a weak La Niña combined with record-high sea temperatures around New Zealand sustained the wet pattern. About 40% of sea-level rise comes from warming seawater expanding.

Ocean heat in 2025

23 ZJ

more ocean heat in 2025 than in 2024

200×

the world's annual electricity use

~25 yrs

for surface heat to reach 2,000 meters

Rain on land can temporarily lower sea level: in 2025, heavy rains refilled Australia's Lake Eyre, turning a desert saltpan into a vast inland sea.

Takeaways

The ocean stores the heat

More than 90% of trapped energy goes into the ocean, setting new records.

Warm seas, wet land

Ocean heat feeds atmospheric rivers and storms that flood communities.

Keep watching

Floats and satellites are essential to track ocean heat and sea level.

Key Terms

Ocean heat content

The total heat stored in the ocean's upper layers, the best single indicator of global warming.

Zettajoule

A huge unit of energy equal to 10^{21} joules, used to measure ocean heat.

Profiling float

A robotic instrument that sinks and rises through the ocean measuring temperature and salinity.

Satellite altimetry

Measuring sea level from space by bouncing radar signals off the ocean surface.

Thermal expansion

The increase in water's volume as it warms, which contributes to sea-level rise.

Vocabulary Path · Unlocking Ocean Heat



Activity

Olympic Pools of Heat

Students use comparisons to grasp the scale of ocean heat and connect it to local rain and flooding.

Materials: Calculator; Local rainfall or flood news story; Poster paper

1. Write down the lesson's comparison: 2025's added ocean heat equals heating 28 billion Olympic pools from 20°C to 100°C.
2. Estimate how many Olympic pools it would take to fill your town's area one meter deep. Compare with 28 billion.
3. Draw a diagram showing how ocean heat leads to evaporation, atmospheric rivers, and flooding.
4. Explain why it takes about 25 years for surface heat to reach 2,000 meters.
5. Find a recent heavy-rain event near you and explain how ocean heat might have contributed.

References

Trenberth, K. (2026). NZ is again being soaked this summer: record ocean heat helps explain it. The Conversation. <https://doi.org/10.64628/AA.ep5fpnnfk>